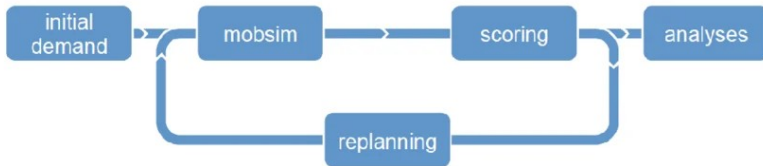


Simple MATSim replanning recipe

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MATSim loop, innovation, and selection



Innovation

creates new travel plans in order to explore behavioral alternatives (and maintain variability in the plan set).

Selection

re-evaluates already existing plans in order to stabilize the assignment and avoid persistent score biases.

Intuition

Only innovation, no score uncertainty

- Agent n with J plans, scores: $S_n^{(1)} \leq \dots \leq S_n^{(J)}$.
- *WorstPlanSelector*: Innovation $S_n^{\text{inno}} \leq \hat{S}_n^{\text{inno}}$ replaces $S_n^{(1)}$ if $S_n^{(1)} < S_n^{\text{inno}}$.
- Sum of scores changes by

$$\max \left\{ 0, S_n^{\text{inno}} - S_n^{(1)} \right\} \geq 0.$$

- Score stops changing once

$$S_n^{(1)} = \dots = S_n^{(J)} = \hat{S}_n^{\text{inno}}.$$

Given no score uncertainty, innovation-only with *WorstPlanSelector* for removal implements a greedy search.

Only innovation, with score uncertainty

Thought experiment

- Two plans A and B with true scores S_A and S_B .
- Innovation draws uniformly over six randomly disturbed possibilities.



- S_A is higher than S_B , but best score of A is lower than best score of B .
- Eventually, B dominates the choice set, although A is better.

Score uncertainty pollutes the score-ranking. Scores in the plan set are upwards-biased. Superior plans may be pushed out of the choice set, while inferior plans may persist.

Innovation and selection, with score uncertainty

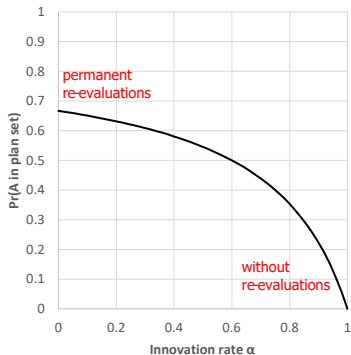
- Innovation with probability α , otherwise plan re-evaluation with *SelectRandom*.
- (*WorstPlanSelector* for removal already implements greedy search.)

Thought experiment continued

- Assume plan set of size $J = 1$.
- Long-term probability of plan A being in the choice set can be evaluated:

$$\Pr(A) = \frac{6(1 - \alpha)}{9 - 7\alpha}.$$

Re-evaluating the plan set counteracts the effect of score uncertainty on the plan ranking, but not entirely.



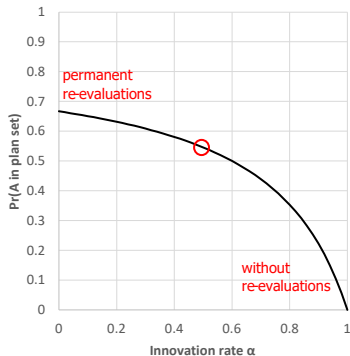
Innovation vs. selection compromise

- Require that each plan is on average selected once between two innovations.

$$\alpha \leq \frac{1}{J+1} \quad \text{for } J \geq 1.$$

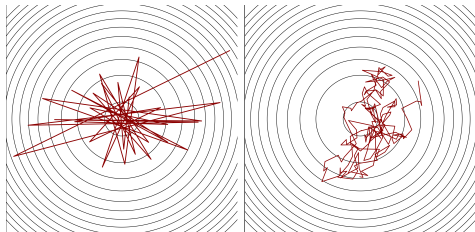
Given that maximum innovation achieves initially fastest progress, start the assignment process with

$$\begin{aligned} J &= 1 \\ \alpha &= 1/2. \end{aligned}$$



Innovation rate size & cooling

- Cooling reduces innovation rate gradually.
- Suitable rate depends on innovation strategies.
- Efficient (left) vs inefficient (right) innovation: →
 - ▶ Inefficient: *TimeAllocationMutator*
 - ▶ Efficient: *ReRoute*
 - ▶ Efficient (few alternatives): *SubtourModeChoice*



- Use Robbins and Monro (1951) conditions to construct cooling schedules.

Cooling should only start after near-stationary conditions have been reached.

- **Slow cooling** if at least one inefficient innovation strategy is used: $\alpha(k) = 1/\sqrt{k+1}$.
- **Fast cooling** if only efficient innovation strategy are used: $\alpha(k) = 1/(k+1)$.

Recipe

Overall settings.

- Use only innovation strategies and *SelectRandom*.
- Do not “turn off” any strategy.
- Limit the plan choice set to one.
- Use *WorstPlanSelector* for plan removal.

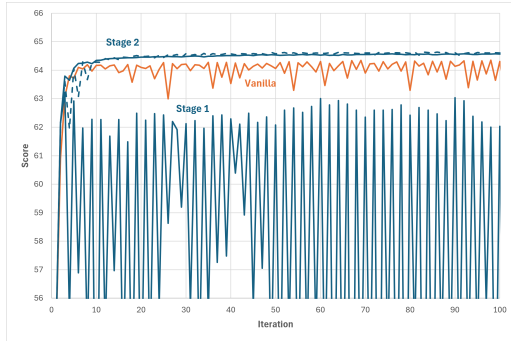
Stage 1. Use a random innovation strategy with probability 0.5. Otherwise, use *SelectRandom*. Run until near-stationarity is attained.

Stage 2. Start with the final plans file from the first stage and with $k = 0$. Use a random innovation strategy with probability $\alpha(k)$ and *SelectRandom* otherwise:

$$\alpha(k) = \begin{cases} \frac{1}{2} \frac{1}{\sqrt{k+1}} & \text{if there is at least one inefficient innovation strategy,} \\ \frac{1}{2} \frac{1}{k+1} & \text{if there are only efficient innovation strategies.} \end{cases}$$

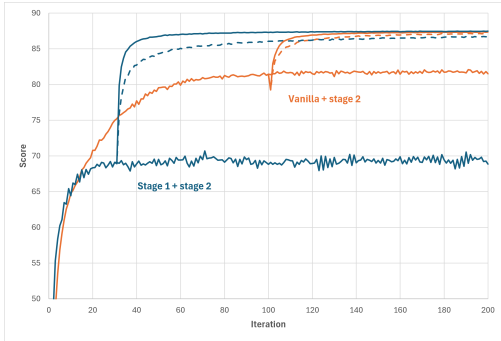
Results

Berlin



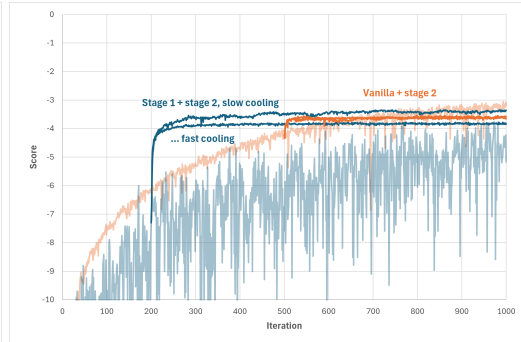
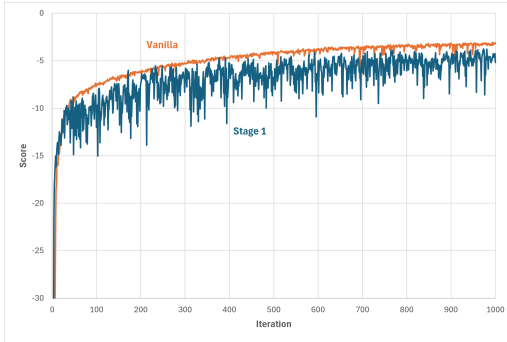
- Innovation: only *ReRoute* → effective.
- Vanilla: $Pr(ChangeExpBeta)=0.9$, five plans per agent.

Munich



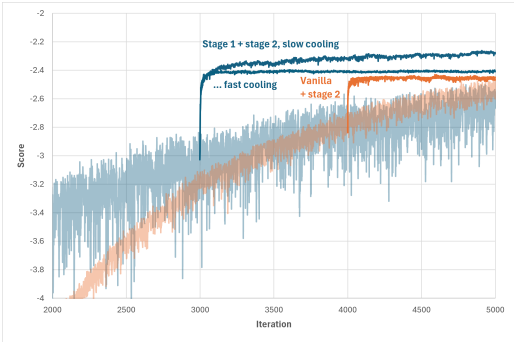
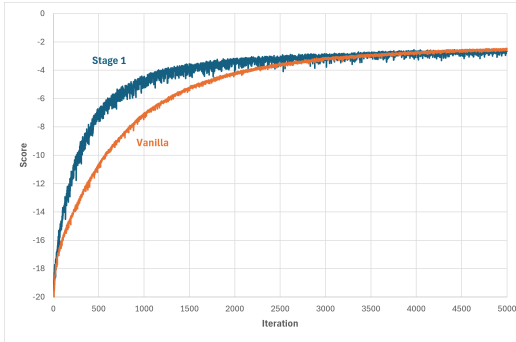
- Innovation: *ReRoute* + *SubtourModeChoice* → effective.
- Vanilla: $\Pr(\text{ChangeExpBeta})=0.8$, four plans per agent.

Stockholm



- Innovation: *ReRoute* + *TimeAllocationMutator* → ineffective.
- Vanilla: $\Pr(\text{ChangeExpBeta})=0.8$, five plans per agent.

Oslo

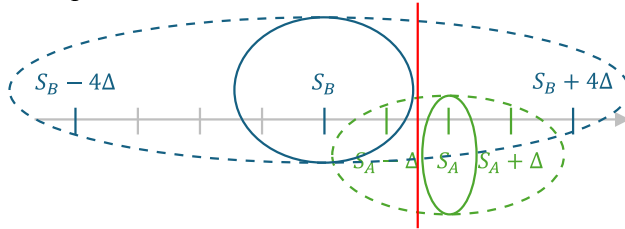


- Innovation: *ReRoute* + *SubtourModeChoice* + *TimeAllocationMutator* → ineffective.
- Vanilla: $\Pr(\text{ChangeExpBeta})=0.7$, five plans per agent.

Low hanging fruits

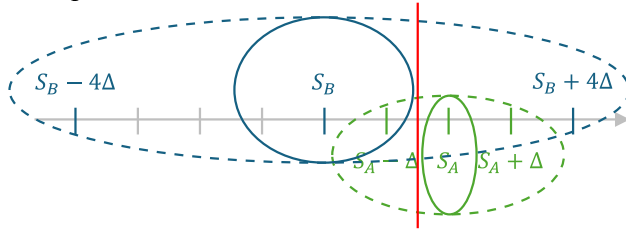
Possibilities for further improvement

- Average scores over iterations.



Possibilities for further improvement

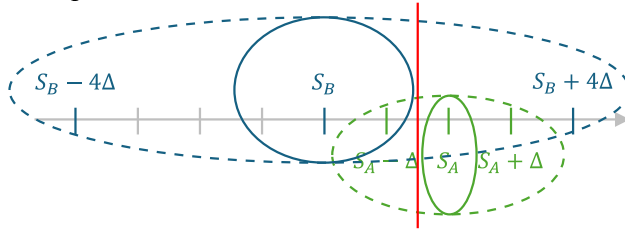
- Average scores over iterations.



- Replace *TimeAllocationMutator* by more directed time allocation.

Possibilities for further improvement

- Average scores over iterations.



- Replace *TimeAllocationMutator* by more directed time allocation.
- Replace co-evolution by a traffic assignment solver → ATAP contrib.